Kanishka Sahoo

Software Development Engineer

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2026 Computer Science graduate and software engineer with a security background. Builds backend services with Python, FastAPI, and PostgreSQL, deployed on AWS and Kubernetes. The internal platform at GoTo was adopted across an engineering organization of 1000+ engineers.

Education

PES University 2022 – 2026

B.Tech. in Computer Science and Engineering; GPA: 8.7/10

- Specialization: Cyber Security and Connected Systems

Experience

GoTo Technologies India Pvt. Ltd. Jan 2026 – Jul 2026

Security Engineering Intern *Bengaluru, India*

- Built a configuration-management REST API with Python, FastAPI, and PostgreSQL for an internal incident platform running on AWS EKS, where the API was used by 15+ microservices.
- Built an in-memory LRU cache with 10-minute expiry and dynamic reloads into the service (95%+ hit rate when warm), so Python and Go consumers picked up config changes with no restarts or downtime.
- Made config writes idempotent and rate-limited updates to once per minute per key per service, so retried or repeated requests could not apply duplicate changes.
- Built rule-based routing for 4,000+ security findings to owning teams, contributing to a reduction in remediation time from 2 months to 16 days.
- Built semantic search over 4,000+ Jira incidents with pgvector and Azure, plus a RAG chatbot on top of it that recommended the correct remediation in 85% of evaluated incidents.
- Fixed bugs in Go microservices, debugged production failures from application and container logs, and recovered Kubernetes deployments.
- Developed automated tests, maintained GitHub Actions CI/CD workflows, participated in code reviews, and wrote technical documentation for the services.

Projects

Linkify – Link Management and Analytics Platform ksahoo.link | Code

- Built and deployed a full-stack link platform, live at ksahoo.link as my own access-controlled instance, using TanStack Start, React, PostgreSQL, and Drizzle ORM with rule-based redirects, per-click analytics, and migrations.
- Designed REST APIs with scoped expiring bearer keys, role-based access, TOTP/passkey authentication, and PostgreSQL-backed rate limiting.

Multi-Agent Workspace for Security Testing Code

- Built a Python/FastAPI and React workspace coordinating specialist agents with execution state persisted in PostgreSQL.
- Implemented WebSocket streaming with persisted-event replay, Docker-isolated long-running processes, pgvector retrieval, and automated pytest tests.

Publications

Forensic Investigation of Google Drive File Sync Artifacts, *IEEE RECCAP 2026* 10.1109/RECCAP69368.2026.11723586

Technical Skills

Languages: Python, C, JavaScript, TypeScript, SQL, Bash, Go
Backend & Data: FastAPI, Flask, REST APIs, PostgreSQL, pgvector, WebSockets, Relational Database Design
Cloud & Infrastructure: AWS (EKS), Docker, Kubernetes, Linux, GitHub Actions, CI/CD, Git
Engineering: API Design, Automated Testing, Debugging, Production Engineering

Achievements and Certifications

Achievements: Top 10 finisher in Terrathon (24 hr hackathon), multiple time top 10 finisher in ISFCR CTF events
Certifications: CompTIA Security+; AI and ML Certification, E&ICT IIT Roorkee (Current)